

Discrete-time control and MDPs

- Definition 1** *A discrete-time control problem is formed of the following:*
- *A control process $(U_t)_{t \in [T]}$ in some $\mathcal{U}$.*
 - *A state process $(X_t^U)_{t \in [T]}$ in $\mathcal{X} \subseteq \mathbb{R}^d$, a function of prior controls.*
 - *An instantaneous cost function $g : [T] \times \mathcal{X} \times \mathcal{U} \rightarrow \mathbb{R}$.*

An optimal control is a $U^* = (U_t^*)$ which minimises

$$J(x, U) = \sum_{t \in [T]} g(t, X_t^U, U_t)$$

where $X_0 = x$.

To solve this, we decompose the problem into minimising each of the ‘remaining-cost’ problems. With

$$J(t, x, U) = \sum_{s \geq t: s \in [T]} g(s, X_s^{t,x,U}, U_s)$$

the remaining cost from time t onwards given $X_t = x$, we note that

$$J(t, x, U) = g(t, x, U_t) + J(t + 1, X_{t+1}^{t,x,U}, U)$$

and with $V(t, x) := \inf_U J(t, x, U)$, it is clear by the independence of J from prior decisions and states, and the assumed Markov property of (X_t) , $X_{t+1} = f(t, X_t, U_t)$ that

$$V(t, x) = \inf_{U_t} \left\{ g(t, x, U_t) + V(t + 1, f(t, x, U_t)) \right\}$$

which is the Bellman equation.

Numerical methods and Reinforcement Learning

Continuous Deterministic Control

Continuous Stochastic Control